UNITS & MEASUREMENTS

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. SI BASE UNITS

Quantity	Unit	Dimension
Mass	kilogram (kg)	[M]
Length	metre (m)	[L]
Time	second (s)	[T]
Electric current	ampere (A)	[A]
Temperature	kelvin (K)	[K] or $[\theta]$
Amount of substance	mole (mol)	[mol]
Luminous intensity	candela (cd)	[cd]

Supplementary: plane angle (radian), solid angle (steradian) — dimensionless

Practical units: 1 light year $= 9.46 \times 10^{15}$ m; 1 AU $= 1.496 \times 10^{11}$ m; 1 parsec $= 3.08 \times 10^{16}$ m; 1 Å$= 10^{-10}$ m; 1 fermi $= 10^{-15}$ m

⋆ Common Pitfall

- Radian and steradian have units but **no dimensions**
- Parsec > light year > AU

2. DIMENSIONS OF IMPORTANT QUANTITIES

Quantity	Dimension	Quantity	Dimension
Force	$[MLT^{-2}]$	Pressure, stress, Y	$[ML^{-1}T^{-2}]$
Work, energy, torque	$[ML^2T^{-2}]$	Power	$[ML^2T^{-3}]$
Momentum, impulse	$[MLT^{-1}]$	Ang. momentum, h	$[ML^2T^{-1}]$
Moment of inertia	$[ML^2]$	G	$[M^{-1}L^3T^{-2}]$
Surface tension, k	$[MT^{-2}]$	Viscosity η	$[ML^{-1}T^{-1}]$
Specific heat	$[L^2T^{-2}K^{-1}]$	Latent heat	$[L^2T^{-2}]$
Thermal cond. k	$[MLT^{-3}K^{-1}]$	Stefan σ	$[MT^{-3}K^{-4}]$
R, k_B	$[ML^2T^{-2}K^{-1}]^*$	Charge q	$[AT]$
Potential V	$[ML^2T^{-3}A^{-1}]$	Resistance	$[ML^2T^{-3}A^{-2}]$
Capacitance	$[M^{-1}L^{-2}T^4A^2]$	Inductance	$[ML^2T^{-2}A^{-2}]$
B field	$[MT^{-2}A^{-1}]$	**E** field	$[MLT^{-3}A^{-1}]$
ε_0	$[M^{-1}L^{-3}T^4A^2]$	μ_0	$[MLT^{-2}A^{-2}]$

* R is per mole: $[ML^2T^{-2}K^{-1}\text{mol}^{-1}]$

Handy identities: $\dfrac{1}{\sqrt{\mu_0\varepsilon_0}} = c$ $\quad \dfrac{L}{R} = RC = \sqrt{LC} = [T]$

$\dfrac{E}{B} = [LT^{-1}]$ $\quad \dfrac{B^2}{\mu_0} = \varepsilon_0 E^2 =$ energy density $[ML^{-1}T^{-2}]$

⋆ Common Pitfall

- Same dimension, different quantity: work & torque; stress & pressure & energy density; h & angular momentum
- Arguments of sin, log, e^x must be dimensionless

3. USES OF DIMENSIONAL ANALYSIS

1. Homogeneity check: only like dimensions can be added/equated

2. Unit conversion:

$$n_2 = n_1 \left[\frac{M_1}{M_2}\right]^a \left[\frac{L_1}{L_2}\right]^b \left[\frac{T_1}{T_2}\right]^c$$

3. Deriving relations: assume $Q = k\,x^a y^b z^c$, equate powers of M, L, T

New system of units: if F, v, T are chosen as base, write $M = [F^a V^b T^c]$ and solve

⋆ Common Pitfall

- Cannot find dimensionless constants (2π, $\frac{1}{2}$)
- Fails for relations with sin, log, e^x, or sums of terms
- Fails if the quantity depends on more than 3 variables (in mechanics)

4. SIGNIFICANT FIGURES

- All non-zero digits are significant
- Zeros between non-zero digits are significant: 2005 → 4
- Leading zeros are **not** significant: 0.0052 → 2
- Trailing zeros after decimal **are** significant: 2.500 → 4
- Trailing zeros without decimal: ambiguous — use scientific notation

Addition/Subtraction: result has least number of **decimal places**

Multiplication/Division: result has least number of **significant figures**

Rounding: digit > 5 round up, < 5 drop; exactly 5 → make preceding digit even (2.745 → 2.74, 2.735 → 2.74)

⋆ Common Pitfall

- Change of units does not change significant figures: 4.700 m = 4700 mm (4 s.f.)
- Exact numbers (counted, defined) have infinite significant figures

5. ERRORS IN MEASUREMENT

Mean: $\bar{a} = \dfrac{1}{n}\sum a_i$, $\quad$ Absolute error: $\Delta a_i = |\bar{a} - a_i|$

Mean absolute error: $\overline{\Delta a} = \dfrac{1}{n}\sum |\Delta a_i|$

Relative error $= \dfrac{\overline{\Delta a}}{\bar{a}}$, $\quad$ Percentage error $= \dfrac{\overline{\Delta a}}{\bar{a}} \times 100\%$

Types: systematic (instrumental, personal, imperfect technique), random, least count error

6. PROPAGATION OF ERRORS

Relation	Maximum error
$Z = A \pm B$	$\Delta Z = \Delta A + \Delta B$
$Z = AB$ or A/B	$\dfrac{\Delta Z}{Z} = \dfrac{\Delta A}{A} + \dfrac{\Delta B}{B}$
$Z = A^n$	$\dfrac{\Delta Z}{Z} = n\dfrac{\Delta A}{A}$
$Z = \dfrac{A^p B^q}{C^r}$	$\dfrac{\Delta Z}{Z} = p\dfrac{\Delta A}{A} + q\dfrac{\Delta B}{B} + r\dfrac{\Delta C}{C}$

Example: $g = \dfrac{4\pi^2 L}{T^2} \Rightarrow \dfrac{\Delta g}{g} = \dfrac{\Delta L}{L} + 2\dfrac{\Delta T}{T}$

If T is from n oscillations of total time t: $\Delta T/T = \Delta t/t$

⋆ Common Pitfall

- Errors always **add** — even for subtraction and division
- Quantity with the highest power contributes most error — measure it most precisely
- Absolute error has units; relative error is dimensionless

7. VERNIER CALLIPERS & SCREW GAUGE

Vernier: $\text{LC} = 1\,\text{MSD} - 1\,\text{VSD} = \dfrac{1\ \text{MSD}}{n}$

(n VSD $= (n-1)$ MSD)

Reading = MSR + (VSR × LC) − zero error

Screw gauge: $\text{Pitch} = \dfrac{\text{distance moved}}{\text{no. of rotations}}$

$$\text{LC} = \frac{\text{Pitch}}{\text{no. of circular scale divisions}}$$

Reading = PSR + (CSR × LC) − zero error

Zero error	When	Correction
Positive	zero of VS/CS beyond MS zero	subtract
Negative	zero of VS/CS before MS zero	add (magnitude)

Typical LC: vernier 0.1 mm, screw gauge 0.01 mm

⋆ Common Pitfall

- Corrected reading = observed − (zero error with sign)
- Negative zero error of screw gauge: $(N-x)\times$LC, where x = division coinciding

8. JEE SHORTCUTS & MASTER TIPS

1. Remember: [Energy] = $[ML^2T^{-2}]$, [Pressure] = $[ML^{-1}T^{-2}]$
2. $\sqrt{\mu_0\varepsilon_0} = 1/c$; RC, L/R, $\sqrt{LC}$ are times
3. % error in A^pB^q/C^r: $p\%A + q\%B + r\%C$
4. Answer to $n_1\times$(old unit) = $n_2\times$(new unit) using $[M]^a[L]^b[T]^c$
5. Most precise instrument = smallest least count

⋆ Master Confusion List

1. **Accuracy** = closeness to true value; **precision** = closeness among readings / resolution
2. **Sig. figs:** add/subtract → decimal places; multiply/divide → sig. figs
3. **Errors** always add, never subtract
4. **Torque vs work:** same dimension, different nature (vector vs scalar)
5. **Zero error sign** decides add or subtract

For more resources, visit **letsplaywithphysics.in**